

Renal Reference — Polyuria in AKI

polyuria in AKI

- there is polyurea with AKI and polyurea without AKI in our material

Core idea: why polyuria happens in AKI

Big mechanism	Core idea	What happens
1. Impaired concentrating ability	Tubules/interstitium cannot concentrate urine	Patient loses dilute urine
2. Impaired tubular reabsorption	Injured tubules cannot reabsorb Na ⁺ /water properly	Salt + water stay in urine → high urine output
3. Osmotic diuresis	Solute stays in tubule and pulls water with it	Glucose/urea/mannitol drag water into urine

Differential diagnosis of polyuria in AKI

Cause	Core mechanism	Classic setting	Clues
Post-obstructive diuresis	After obstruction is relieved, kidney dumps retained salt/water/urea + tubules still cannot concentrate urine	After Foley, ureteral stent, <u>nephrostomy</u> , relief of BPH/stone obstruction	Very high urine output after obstruction relieved; risk of dehydration, hypokalemia, hypotension
<u>Tubulointerstitial disease</u> / AIN	Interstitial inflammation damages tubular concentrating ability	Drug-induced AIN: PPIs, NSAIDs, antibiotics, checkpoint inhibitors	WBCs, WBC casts, sterile pyuria; fever/rash/eosinophilia may occur but often absent
Nephrotoxic <u>nonoliguric</u> ATN	Tubular injury but urine flow preserved or increased	<u>Aminoglycosides</u> , cisplatin, amphotericin B, ifosfamide, tenofovir	<u>Nonoliguric</u> AKI; granular casts/RTECs; drug exposure; <u>aminoglycosides</u> are classic
Recovery phase of ATN	GFR starts recovering before tubules fully recover reabsorption/concentration	After ischemic or toxic ATN begins improving	Creatinine stabilizes/improves, then polyuria; risk of volume/electrolyte losses
Osmotic diuresis	Solute in tubular lumen pulls water into urine	Hyperglycemia/DKA/HHS, mannitol, high urea load	Glucosuria or high solute load; can cause volume depletion → <u>prerenal</u> AKI → ATN

1. Two forms exist

WHAT YOU SEE

There is polyuria WITH AKI and polyuria WITHOUT AKI in our material

WHAT IT ENCODES

Frames the topic: AKI is NOT always oliguric. Polyuria is defined as urine output >3 L/day (or >3.5 L/24h). Non-oliguric AKI (urine >400–500 mL/day with a rising creatinine) carries a better prognosis and lower dialysis need than oliguric AKI, but the elevated output can mask ongoing injury and drive dangerous volume/electrolyte losses.

BOARD TRAP

WRONG: 'good urine output rules out AKI, so no need to recheck creatinine.' Non-oliguric/polyuric AKI exists — the kidney makes plenty of dilute urine but still fails to clear solute. Discriminator: a rising creatinine despite brisk urine output is still AKI; trend the creatinine, not just the output.

2. Mechanism 1 — impaired concentration

WHAT YOU SEE

Impaired concentrating ability: tubules/interstitium cannot concentrate urine → patient loses dilute urine

WHAT IT ENCODES

Damaged tubules/interstitium cannot reabsorb free water against the medullary osmotic gradient (loss of the countercurrent mechanism and/or ADH responsiveness), so the patient excretes inappropriately dilute urine. Urine osmolality is low/iso-osmotic (~300 mOsm/kg) despite a need to concentrate. Seen in tubulointerstitial disease, recovery-phase ATN, and washout of the medullary gradient.

BOARD TRAP

WRONG: assuming low urine osm with polyuria means the patient simply drank too much. A concentrating defect from tubular injury produces dilute urine the kidney CANNOT correct, unlike primary polydipsia. Discriminator: in a damaged kidney the urine stays dilute even when the patient is volume-down/hyponatremic; in polydipsia it concentrates once water is restricted.

3. Mechanism 2 — impaired Na/water reabsorption

WHAT YOU SEE

Impaired tubular reabsorption: injured tubules cannot reabsorb Na⁺/water → salt + water stay in tubule → high urine output

WHAT IT ENCODES

Injured proximal/distal tubules cannot reabsorb filtered Na⁺ and water, so salt and water stay in the lumen and exit as urine. This is a SALT-wasting polyuria (high urine Na, FENa elevated) — contrast with the free-water diuresis of mechanism 1. It contributes to the recovery (polyuric) phase of ATN and post-obstructive diuresis.

BOARD TRAP

WRONG: replacing these losses with free water (D5W) only. The loss is iso-osmotic salt + water, so pure water replacement causes hyponatremia while volume keeps dropping. Discriminator: check urine Na/osm — a high-Na, iso-osmotic diuresis needs isotonic crystalloid replacement, not just water.

4. Mechanism 3 — osmotic diuresis

WHAT YOU SEE

Osmotic diuresis: solute stays in tubule and pulls water with it (glucose/urea/mannitol)

WHAT IT ENCODES

Non-reabsorbed solute (glucose once filtered load exceeds the ~180–200 mg/dL threshold, urea from a high catabolic/protein load or recovering ATN, mannitol) holds water osmotically in the tubular lumen. Urine osm is paradoxically not minimized (solute present) and the diuresis is rich in both water and solute. Spot urine osm with calculated electrolyte-free water clearance separates osmotic from water diuresis.

BOARD TRAP

WRONG: calling DKA-related high urine output a sign the patient is well-hydrated. Glucose/urea/mannitol drag water out osmotically and cause profound volume depletion that can precipitate prerenal AKI → ATN. Discriminator: glucosuria/high urea load with polyuria means the patient is being dried out, not over-hydrated — replace volume.

5. Post-obstructive diuresis

WHAT YOU SEE

DDx: Post-obstructive diuresis — kidney dumps retained salt/water + tubules still can't concentrate

WHAT IT ENCODES

After relief of obstruction (Foley for BPH, ureteral stent, percutaneous nephrostomy, passing a stone) the kidney dumps retained salt, water, and urea (appropriate physiologic diuresis) compounded by a transient tubular concentrating/reabsorptive defect (pathologic component). Output can exceed 200 mL/h. Most diureses are appropriate and self-limited; replace only ~half to two-thirds of urine output with isotonic fluid to avoid perpetuating it.

BOARD TRAP

WRONG: matching urine output mL-for-mL with IV fluid after relieving obstruction. Aggressive 1:1 replacement perpetuates a diuresis that would otherwise self-limit. Discriminator: distinguish appropriate (resolving on its own as retained solute clears) from pathologic (true tubular defect) — replace partially, monitor weight, Na, K, and volume status.

6. Tubulointerstitial disease / AIN

WHAT YOU SEE

DDx: Tubulointerstitial disease / AIN — interstitial inflammation damages concentrating ability

WHAT IT ENCODES

Interstitial inflammation injures the medullary architecture and tubular ADH responsiveness → a concentrating defect with polyuria. Drug-induced AIN is the leading cause: PPIs, NSAIDs, beta-lactams/antibiotics, sulfa, allopurinol, and immune checkpoint inhibitors. Sediment shows WBCs, WBC casts, and sterile pyuria. First step is stopping the culprit drug; steroids if no recovery in ~5–7 days.

BOARD TRAP

WRONG: requiring fever + rash + eosinophilia to call AIN. The classic triad is present in well under 10–15% of cases and is usually INCOMPLETE/absent, especially with PPIs and checkpoint inhibitors. Discriminator: subacute AKI + sterile pyuria + WBC casts + a new drug is enough — eosinophiluria is neither sensitive nor specific.

7. Nephrotoxic nonoliguric ATN

WHAT YOU SEE

DDx: Nephrotoxic nonoliguric ATN — tubular injury, urine flow preserved/increased

WHAT IT ENCODES

Direct tubular toxins injure cells while urine flow stays preserved or increased (non-oliguric ATN). Classic agents: aminoglycosides (non-oliguric, with Mg/K wasting, ~5–10 days in), cisplatin (Mg wasting), amphotericin B (distal RTA + K/Mg wasting), ifosfamide (Fanconi), tenofovir (proximal/Fanconi). Sediment shows muddy-brown granular casts and renal tubular epithelial cells.

BOARD TRAP

WRONG: 'urine output is high, so the tubules are fine.' Non-oliguric ATN means tubular necrosis WITH preserved output — high flow does not rule out injury. Discriminator: granular casts/RTE cells + a nephrotoxin + rising creatinine confirm ATN despite robust urine output.

8. Recovery phase of ATN

WHAT YOU SEE

DDx: Recovery phase of ATN — GFR recovers before tubules recover reabsorption

WHAT IT ENCODES

During recovery from ischemic/toxic ATN the GFR (filtration) returns before the tubules regain reabsorptive/concentrating function, plus accumulated urea/solute drives an osmotic component → a brisk polyuric phase. Creatinine stabilizes then falls as urine output climbs. This phase carries real risk of hypovolemia, hypokalemia, and hypophosphatemia.

BOARD TRAP

WRONG: relaxing monitoring once the creatinine starts improving. The polyuric recovery phase can cause life-threatening volume and electrolyte (K, Mg, PO₄) depletion right when everyone assumes the patient is better. Discriminator: rising output with a falling creatinine after ATN = recovery diuresis — keep replacing fluids/electrolytes and tracking labs.

9. Osmotic diuresis (DDx)

WHAT YOU SEE

DDx: Osmotic diuresis — solute in tubular lumen pulls water into urine

WHAT IT ENCODES

Same mechanism as mechanism 3, now listed as a discrete differential: hyperglycemia (DKA/HHS), mannitol, and a high urea load (catabolism, GI bleed, high-protein/TPN feeds, recovering ATN) each create an obligate solute diuresis. Urine osm is not maximally dilute and total daily solute excretion is high.

BOARD TRAP

WRONG: treating osmotic-diuresis polyuria as a primary water-handling defect (e.g., chasing DI). The driver is solute, not ADH; if severe it causes volume depletion → prerenal AKI → ATN. Discriminator: a high urine osm/solute excretion with an obvious solute source (glucose, mannitol, urea) points to osmotic diuresis, not central/nephrogenic DI.